# Moving towards a semantic archival edition: the PAVES-e project

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## Introduction

The paper presents the open-access semantic edition-archive defined within the PAVES-e project (D'Agata et al. 2024)[1]. The DigitalPavese web environment, specially designed and developed according to the FAIR (Wilkinson et al. 2016) and TRUST (Lin et al. 2020) principles, provides access to the literary and documentary heritage of Cesare Pavese (1908-1950), Italian writer, poet and intellectual.

From a theoretical point of view, the PAVES-e project adopts a 'holistic' approach to represent data by integrating different perspectives of the text. In particular, the project takes into account: 1) the archival nature, guaranteed by the informative exhaustiveness and objective verification of the documents; 2) the critical-textual dimension, which emerges from the interpretative and subjective expertise of the editor; 3) the factual and functional aspect, formally represented through the explication of the semantics of the data.

## Knowledge and Data Organisation

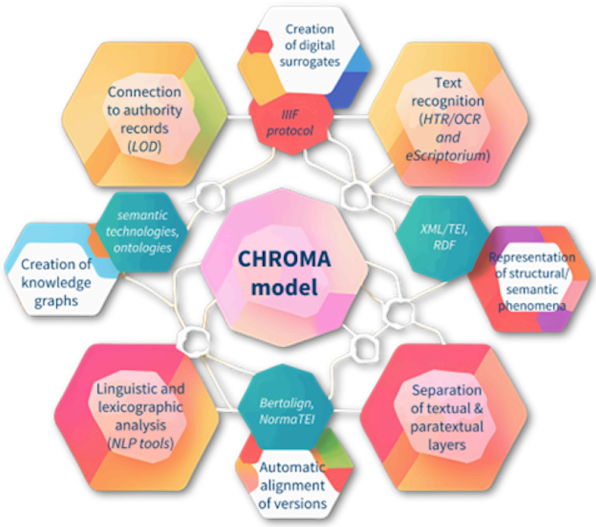
This approach is applied to a corpus that includes Pavese's poetic works, novels, the *Dialoghi con Leucò*, correspondence, essays, and diary, as well as secondary literature. The works are annotated in accordance with the TEI Guidelines and integrated into a knowledge graph representing the entire corpus, aligning with the principles of knowledge organization in the relevant literature (Eggert 2019, Tomasi 2020, Vogeler 2019).

An exciting and challenging aspect of the project lies in the highly varied nature of the data and the numerous related properties and relationships (i.e., the *semantic knowledge base* : Allemang et al. 2020) to consider. In fact, a formal ontology— *OntoPavese*[2]—is being developed, providing a comprehensive framework for the *semantic knowledge organization* within the project. OntoPavese is a highly expressive ontology that covers a wide range of cases in terms of representation and description of Pavese's production items, their (hierarchical) nature, and relationships, and of several other related entities.

## Workflow

The process of creating, preserving, processing, publishing, and querying concerning complex textual resources requires a methodological framework that is both broadly applicable and domain-oriented. While its generic design facilitates adoption across diverse contexts, domain features ensure effective implementation. To meet these needs, the CHROMA model (Cristofaro et al. 2023, Sichera et al. 2024) has been proposed (figure 1). The model, already applied in previous projects such as *BDC* (Cristofaro et al. 2023, Cristofaro et al. 2025, Del Grosso et al. 2023)[3], *COVerLeSS*[4] and *Pirandello Nazionale* (D'Agata et al. 2022, Sichera et al. 2019)[5], envisages text as a complex, multidimensional object managed through an editorial process[6]. The CHROMA model encompasses a series of activities defined through a double articulation: representation formats on one

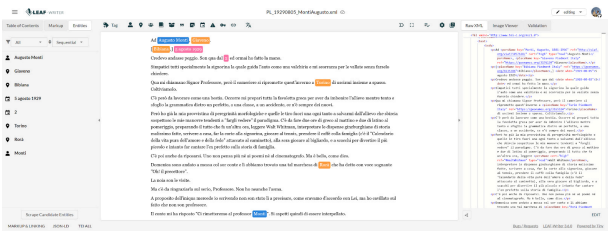
hand, and multiple tasks on the other. Specifically, the model makes available the digital surrogate of the primary source (IIIF), and represents structural and semantic phenomena using generic languages like XML/TEI and RDF or domain-specific languages (DSLs). Encoding is supported by harmonization strategies[7], integrating linguistic and lexicographic analyses. These data gain significance through knowledge graphs, linked to authority records via LOD. The edition is further enriched with interactive visualization tools that enhance accessibility and user engagement.




*Diagrammatic representation of the CHROMA model.*

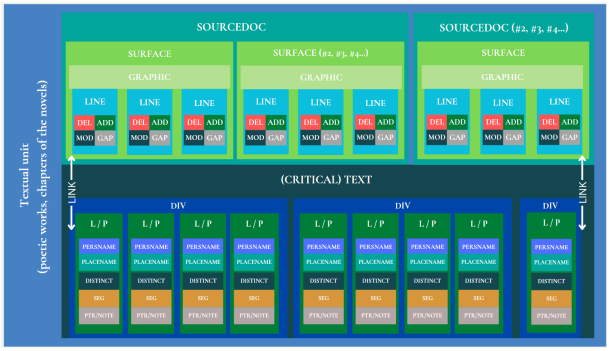
# Text Encoding

The text encoding model follows the schema defined and promoted by the *TEI* (TEI Consortium 2024) and was developed based on the fundamental distinction between creative works and archival documents. Based on this differentiation, named entities were encoded for archival documents, along with technical terms or unusual words, using the open-source platform LEAF (figure 2).



*LEAF interface displaying the encoding of a letter.*

Texts were subsequently revised to mark their structure and add notes through a different editor: the text of the critical edition was therefore placed within the <body> in the <text> element, encoding the structure, named entities, lexicon, and notes. For creative works, the documentary dimension of the manuscripts was also represented, through various <sourceDoc> elements, each containing one or more <surface> elements. Within these, handwritten lines were marked with the <line> tag, while corrective phenomena were encoded with <del>, <add>, <mod>, <delSpan>. The various levels of encoding are summarized in figure 3, and an example of the encoding of a witness is provided in figure 4.




*Representation of levels in <sourceDoc> and <text>.*

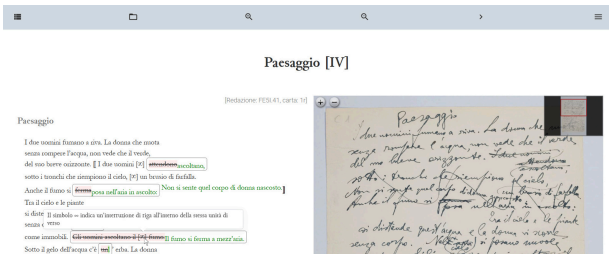
```
<sourceDoc xml:id="LS_PaesaggioIV_FE5I.41_A" n="A">
  <pb n="1" xml:id="FE5I.41_1r"/>
  <surface n="1r">
    <graphic url="LS_FE5I.41_1r.png"/>
    <line type="head">Paesaggio</line>
    <line>I due uomini fumano a riva. La donna che nuota</line>
    <line>senza rompere l'acqua, non vede che il verde,</line>
    <line>del suo breve orizzonte.<delSpan spanTo="#delSpan_PaesaggioIV_1"/> I due uomini<lb/>
      <mod type="subst">
        <del>attendono</del>
        <add place="below">ascoltano,</add>
      </mod>
    </line>
    <line>sotto i tronchi che riempiono il cielo,<lb/> un brusio di farfalla.</line>
    <line>Anche il fumo si
      <mod type="subst">
        <del>ferma</del>
        <add place="below">posa nell'aria in ascolto:</add>
      </mod>
      <add place="above" type="alternative">Non si sente quel corpo di donna nascosto.</add>
      <anchor xml:id="delSpan_PaesaggioIV_1"/>
    </line>
```

*Encoding of surface 1r of the FE5I.41 witness (poem "Paesaggio [IV]" from "Lavorare stanca", 1936).*

# Visualizing and Analyzing Pavese's Works

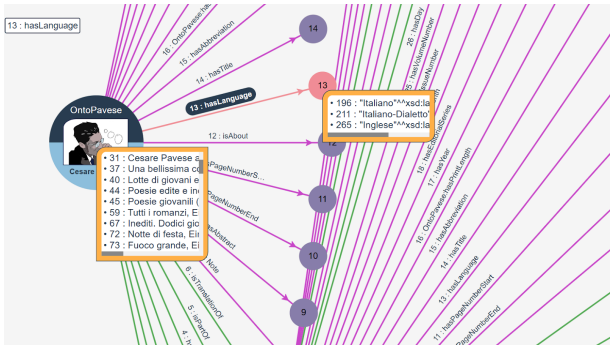
The annotation of texts is an essential prerequisite for ensuring the visualization and consultation of the edition on the web, which is being developed within the framework of *PaveseInTesto*, a

digital platform developed using the open-source software TEI Publisher[8]. TEI Publisher enables the creation of scholarly digital editions of Pavese's works with maximum adaptability to the unique features of each textual typology, without imposing the constraints of a 'one-size-fits-all' framework. Specifically, for archival documents, a full-text layout has been designed, allowing access to comments and explanatory notes, chronological connections between contemporary works by the author, intertextual citations, bibliographic references, and a list of named entities (referred to as the 'text and comment' mode). For the edition of creative works, additional features include a viewing mode that combines facsimiles with 'diplomatic' transcriptions (figure 5)[9] and a 'collation' mode, which allows users to juxtapose various drafts of the same textual unit with the critical edition in a customizable order.




*TEI Publisher interface for the 'facsimile + diplomatic transcription' viewing mode.*

In regard to the *OntoPavese ontology*, it is in plan the development (currently in progress) of a web tool— *OntoPavesePathExplorer*—that allows the visual exploration of ( *ontological*) *paths* generated by *property chains*, where a path is, in the present context, a sequence of class instances (the *nodes* of the path) each of which is connected to the subsequent one by means of an Object property assertion. Thus, after (visually) selecting a *target instance* of interest, one can explore all paths ending/starting at that instance; this provides an easy and intuitive mechanism for accessing useful information about the semantic organization of the ontology at the level of instance relationships—information that otherwise could/should be retrieved (e.g.) by means of alternative ontology-query interfaces exploiting the "not-so-easy" SPARQL query language (or the likes). The visual interface of *OntoPavesePathExplorer* (figure 6) will integrate with that of *PaveseInTesto*, working in synergy with it, and allowing on-demand switching between the views offered by both.




*The visual interface of (an early prototype version of) OntoPavesePathExplorer.*

# Further development

Pavese's oeuvre stands out as a rare instance of a twentieth-century author who engaged comprehensively across almost every literary form, from poetry to the novel, culminating in the unique *Dialoghi con Leucò*. His corpus, which also includes personal (and editorially relevant) correspondence and a diary, provides a rich dataset for analyzing authorial practices and editorial dynamics in the early twentieth century. In the coming months, further development is planned for the lemmatization of the works. Specifically, the project aims to implement an integrated system that will combine concordances, TEI encoding, and the use of lexical ontologies such as BabelNet[10] under a single processing and querying platform.

# Notes

1. https://digitalpavese.cnr.it/ (funded by the European Union – Next Generation EU, Mission 4 Component 2, Investment 1.1, CUP E53D23013670006).
2. https://digitalpavese.cnr.it/ontopavese/
3. https://bellinicorrespondence.cnr.it/
4. https://coverless.cnr.it/ ("Corpus Online del Verismo tra Letteratura, Storia e Società", funded by the European Union – Next Generation EU, Mission 4 Component 1, Investiment 1.1, CUP E53D23018880001).
5. https://www.pirandellonazionale.it/
6. See also the *Verismo Digitale* project, funded by the European Union – NextGenerationEU (Project PE 0000020 CHANGES – CUP E63C22001960006, PNRR Mission 4 Component 2 Investment 1.3), aimed at creating a digital ecosystem around the works of Verismo literature.
7. See, for example, NormaTEI (Sichera 2024), a harmonization tool developed on the *BDC* edi-

tion and further expanded within the framework of the H2IOSC Project – Humanities and Cultural Heritage Italian Open Science Cloud, funded by the European Union – NextGenerationEU – NRRP M4C2, Project code IR0000029 – CUP B63C22000730005.
8. https://teipublisher.com/
9. A prototype of the 'facsimile + diplomatic transcription' viewing mode for all the poems in *Lavorare stanca* is available at https://digital-pavese.cnr.it/poesie/ (accessible via the "Vai all'edizione" button in the "Edizione diplomatica" section).
10. https://babelnet.org/

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